Section 14.2: Limits and Continuity

Partial Derivatives

Limits in Multiple Variables

In Calc I, we asked: what happens to $f(x)$ as $x \to a$?

There were only **two directions** to approach from — left and right.

In Calc III, we ask: what happens to $f(x, y)$ as $(x, y) \to (a, b)$?

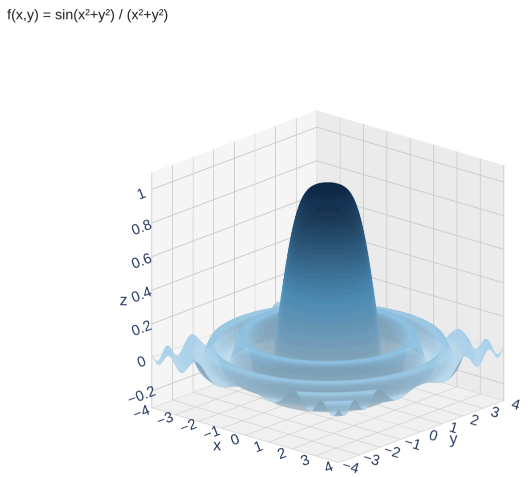
Now there are **infinitely many paths** to approach from — along lines, curves, spirals, any path you can imagine.

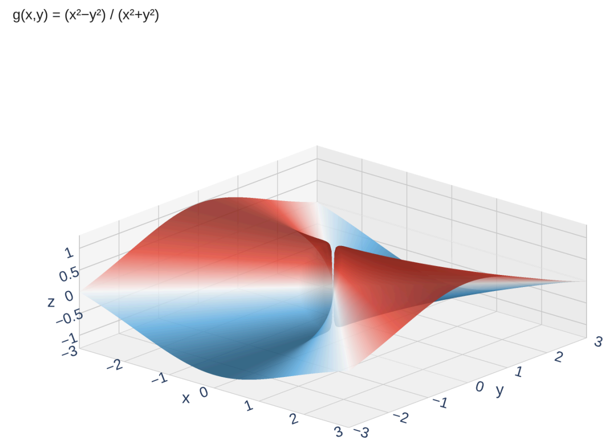
> This is the key difference: in 1D you check two directions. In 2D you must worry about **every possible path** to the point.

Two Functions, Two Very Different Behaviors

Let's compare what happens at the origin for two functions that both have the form $\frac{0}{0}$ there.

$$f(x,y) = \frac{\sin(x^2+y^2)}{x^2+y^2} \qquad\qquad g(x,y) = \frac{x^2-y^2}{x^2+y^2}$$





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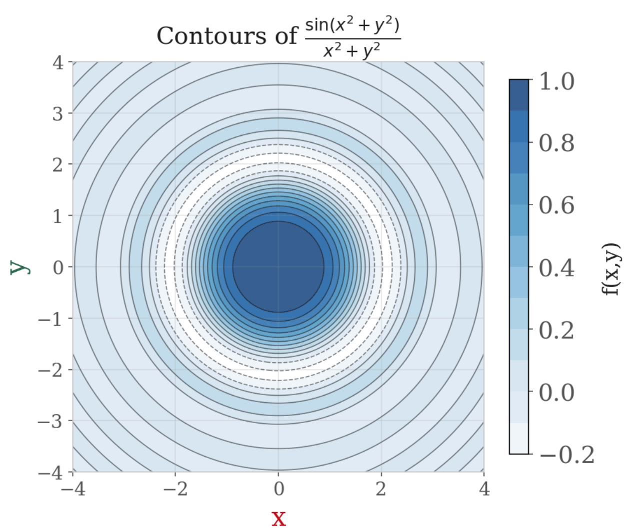

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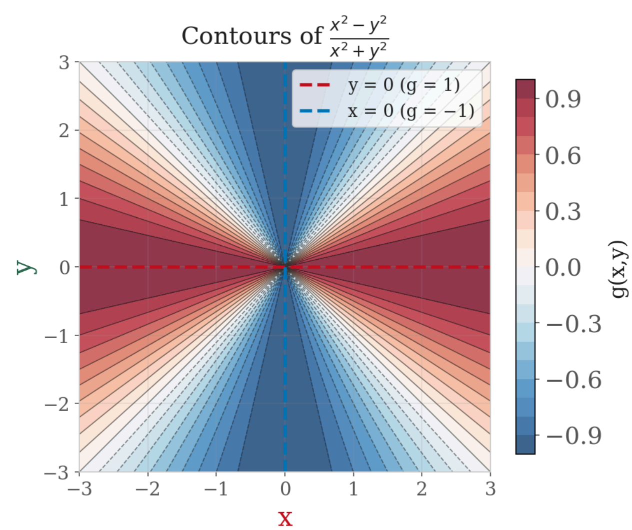

f settles down to a single value (1) no matter how you approach the origin.

g gives **different values** depending on your path — the limit **does not exist**.

The Formal Definition of a Multivariable Limit

Limit of a Function of Two Variables

Let f be a function of two variables whose domain D includes points arbitrarily close to (a,b). We say that the **limit of $f(x,y)$ as (x,y) approaches (a,b)** is L, and write

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

if for every $\varepsilon > 0$ there is a corresponding $\delta > 0$ such that

$$\text{if } (x,y) \in D \text{ and } 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \text{ then } |f(x,y) - L| < \varepsilon$$

What does this mean in plain language?

If the limit is going to be L, then you must be able to make $f(x,y)$ as close to L as you want, just by squeezing (x,y) close enough to (a,b) — from **any direction**.

The crucial phrase: **from any direction**. In Calc I, "closeness" meant $|x-a| < \delta$ (an interval). Here, closeness means $\sqrt{(x-a)^2 + (y-b)^2} < \delta$ (a disk).

In practice, we rarely use the ε-δ definition directly. Instead, we develop **strategies** for showing limits exist or don't exist.

Strategy: Showing a Limit Does NOT Exist

If a limit exists, it must be the **same value** no matter which path you take to the point. So to show a limit **does not** exist, we just need to find trouble:

The Two-Path Test

Find two paths to (a,b) that give **different limiting values**. Then the limit does not exist.

Common paths to try:

- Along the x-axis: set $y = 0$
- Along the y-axis: set $x = 0$
- Along $y = x$ (diagonal)
- Along $y = mx$ (any line through the origin)
- Along curves: $y = x^2$, $y = x^3$, etc.

Warning: If two paths give the **same** value, that does **not** prove the limit exists. It just means those two paths agree. You would need to check **all** paths.

Example 1: The Two-Path Test in Action

Example 1

Show that $\lim_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Try the paths y = 0 and x = 0. What value does the function approach along each?

Path 1 — Along the x-axis (set $y = 0$):

$$f(x,0) = \frac{x^2 - 0}{x^2 + 0} = \frac{x^2}{x^2} = 1 \quad \Rightarrow \quad \lim_{x\to 0} f(x,0) = 1$$

Path 2 — Along the y-axis (set $x = 0$):

$$f(0,y) = \frac{0 - y^2}{0 + y^2} = \frac{-y^2}{y^2} = -1 \quad \Rightarrow \quad \lim_{y\to 0} f(0,y) = -1$$

Since $1 \neq -1$, the two paths give different values. **The limit does not exist.**

Example 2: When Simple Paths Aren't Enough

Example 2

Show that $\lim_{(x,y)\to(0,0)} \frac{8x^3y}{x^6+y^2}$ does not exist.

Try the obvious paths (axes, y = mx) first, then try a curved path like y = x-cubed.

Let's try the obvious paths first.

Along the x-axis ($y=0$): $\frac{8x^3 \cdot 0}{x^6+0} = 0$

Along the y-axis ($x=0$): $\frac{0}{0+y^2} = 0$

Along $y = mx$: $\frac{8x^3(mx)}{x^6+m^2x^2} = \frac{8mx^2}{x^4+m^2} \to 0$

Every line through the origin gives 0. But matching along all lines is **still not enough**. We need **curved** paths.

Along $y = x^3$:

$$\frac{8x^3 \cdot x^3}{x^6+(x^3)^2} = \frac{8x^6}{2x^6} = 4$$

Along $y = x^3$, the function is **constantly 4**. Since the linear paths gave 0 and the cubic path gives 4: **the limit does not exist.**

Your Turn: Test This Limit

Example 3

Does $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ exist?

Try a path that will obviously give a limit of zero, then see if you can think of path that could do something else.

Along $y = 0$:

$$\frac{x \cdot 0}{x^2 + 0} = 0 \quad \Rightarrow \quad \text{limit} = 0$$

Along $y = x$:

$$\frac{x \cdot x}{x^2 + x^2} = \frac{x^2}{2x^2} = \frac{1}{2} \quad \Rightarrow \quad \text{limit} = \frac{1}{2}$$

Since $0 \neq \frac{1}{2}$, **the limit does not exist**.

Continuity of Multivariable Functions

Continuity works just like Calc I — the limit equals the function value at the point.

Continuity at a Point

A function f of two variables is **continuous at** (a, b) if:

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$$

This requires three things (same as Calc I):

1. $f(a, b)$ is defined
2. The limit exists
3. They are equal

Good news: Polynomials, rational functions (where defined), trig functions, exponentials, logarithms, and compositions of continuous functions are all continuous on their domains. Most functions you encounter are continuous wherever they're defined.

Strategy: Showing a Limit DOES Exist

The two-path test can **disprove** a limit, but it can never **prove** one.

To prove a limit exists, we need tools that handle **all paths at once**.

Tool 1: Squeeze Theorem

If $|f(x, y) - L| \leq g(x, y)$ and $g(x, y) \to 0$ as $(x, y) \to (a, b)$, then $f(x, y) \to L$.

Trap the function between bounds that both go to the same value.

Tool 2: Polar Coordinates

Substitute $x = r\cos\theta$, $y = r\sin\theta$. Then $(x, y) \to (0, 0)$ becomes $r \to 0^+$.

Apply the Squeeze Theorem to bound the θ-dependent factor and show the expression goes to zero.

Both tools convert the multivariable problem into something we can handle with Calc I techniques.

Example 4: Proving a Limit Exists

Example 4

Show that $\lim_{(x,y)\to(0,0)} \frac{5xy^2}{x^2+y^2} = 0$.

Try the Squeeze Theorem using $y^2 \le x^2 + y^2$. Then try polar coordinates and apply the Squeeze Theorem to the result.

Method 1 — Direct Squeeze Theorem:

Since $y^2 \le x^2 + y^2$, we have $\frac{y^2}{x^2+y^2} \le 1$. Therefore:

$$0 \le \left| \frac{5xy^2}{x^2+y^2} \right| = 5|x| \cdot \frac{y^2}{x^2+y^2} \le 5|x| \to 0$$

By the Squeeze Theorem, $\frac{5xy^2}{x^2+y^2} \to 0$. $\square$

Method 2 — Polar Coordinates + Squeeze Theorem:

Let $x = r\cos\theta$, $y = r\sin\theta$:

$$\frac{5xy^2}{x^2+y^2} = \frac{5r^3\cos\theta\sin^2\theta}{r^2} = 5r\cos\theta\sin^2\theta$$

Now squeeze: since $|\cos\theta\sin^2\theta| \le 1$ for all θ:

$$0 \le |5r\cos\theta\sin^2\theta| \le 5r \to 0 \text{ as } r \to 0^+$$

By the Squeeze Theorem, the limit is 0, regardless of path. $\square$

Both methods use the Squeeze Theorem at the end. Polar coordinates simplify the algebra first — the r^2 cancels, leaving a single factor of r to squeeze.

Your Turn: Prove or Disprove

Example 5

Does $\lim_{(x,y)\to(0,0)} \dfrac{x^2y}{x^2+y^2}$ exist? If so, find its value.

Use either the squeeze theorem or polar coordinates.

Polar coordinates: Let $x = r\cos\theta$, $y = r\sin\theta$:

$$\frac{x^2y}{x^2+y^2} = \frac{r^2\cos^2\theta \cdot r\sin\theta}{r^2} = r\cos^2\theta\sin\theta$$

Since $|\cos^2\theta\sin\theta| \le 1$, we get $|r\cos^2\theta\sin\theta| \le r \to 0$. **The limit is 0.**

Homework

Section 14.2: 5, 7, 9, 15, 17, 21, 22, 23, 25, 29, 41, 43, 47, 49, 51